TEXORA · LECTURE SHEET

Pigment Dyeing

Padding insoluble colour with a binder: depth set by pick-up, levelness won or lost in drying, permanence created in the curing oven.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 4 · LEVEL 3 TERM 1

1. The principle — padding colour that has no affinity

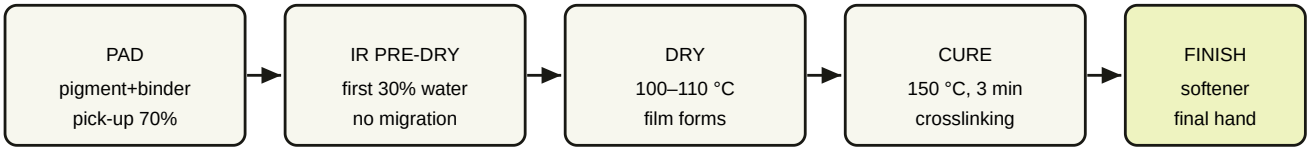
Pigment dyeing (more exactly pigment padding) colours fabric by padding a dispersion of insoluble pigment with a **binder**, drying, and curing so the binder crosslinks into a film that mechanically locks pigment to every fibre surface. Because there is no exhaustion, no fibre chemistry and no wash-off, the process is short, water-lean and works identically on all fibres and blends. The price is paid in handle (the film stiffens), in rub fastness (colour is at the surface) and in depth (heavy pigment loads look dusty). Typical uses: garment dyeing and vintage/washed looks, chambray and denim effects, pastel piece shades, and fabrics that cannot be wet-processed conventionally.

2. The pad liquor — components and roles

COMPONENT	ROLE AND TYPICAL DOSE
Pigment dispersion	Colour; 1–30 g/L by depth (organic pigments, pre-dispersed with wetting/dispersing agents)
Binder (acrylic or butadiene copolymer)	Film former; roughly 1–2× the pigment weight for pale shades, at least equal weight for deep; self-crosslinking grades cure without catalyst
Fixing / crosslinking agent	Melamine-formaldehyde or blocked isocyanate types raise wash and rub fastness
Antimigrant	Alginate/polyacrylate thickener (raises liquor viscosity) so pigment does not migrate during drying
Softener / hand modifier	Silicone or polyethylene emulsion restoring part of the lost softness
Wetting agent, defoamer, ammonia	Rapid wetting, foam control, pH ≈ 8–9 for emulsion stability

3. Process sequence and the physics of each step

1. **Padding:** fabric passes through the liquor and nip rollers at a controlled **wet pick-up** (typically 60–80%). Pick-up, not bath exhaustion, sets the depth: depth ∝ pick-up × liquor concentration.
2. **Drying:** the danger step. As water evaporates from yarn intersections, liquor — and pigment — is dragged to evaporation fronts (**migration**), giving two-sidedness and listing. Defences: antimigrant, infrared pre-drying to remove the first ~30% of water without contact, then hot-flue drying near 100–110 °C.
3. **Curing:** 140–160 °C for 2–4 min crosslinks the binder. Under-cure fails wash/rub tests; over-cure yellows and embrittles the film.



Pigment padding flow — depth is set at the padder, permanence at the curing oven

Fig. 1 — Pigment dyeing (pad–dry–cure) sequence with the migration-control step.

4. A worked pick-up calculation

DEPTH FROM PICK-UP

Liquor contains pigment 12 g/L. Fabric 250 g/m, padded at 70% wet pick-up.
Liquor carried per kg fabric = 0.70 L (taking liquor density ≈ 1 kg/L).
Pigment deposited = 12 g/L × 0.70 L/kg = 8.4 g per kg = **0.84% owf**.
To deepen to 1.68% owf at the same pick-up, double the liquor to 24 g/L — or raise pick-up, which also raises binder pick-up and handle loss. This linear arithmetic is why pigment padding is so reproducible.

5. Advantages, limitations, and garment dyeing

- **Advantages:** any fibre/blend, uniform on blends; no salt, no wash-off (low water/energy); excellent light fastness of inorganic and phthalocyanine pigments; easy shade correction at the padder.
- **Limitations:** dry rub fastness often grade 2–3 in deep shades; handle stiffer; depth ceiling before the shade turns dusty; film can crack on stretch fabrics.

- **Garment pigment dyeing** uses exhaust-style machines with cationised cotton or special binders to give washed-down vintage effects — here unevenness is the product, deliberately produced and then fixed with binder in a second step.

6. Glossary — key terms

- **Wet pick-up:** liquor retained after padding, as % of dry fabric weight.
- **Migration:** movement of colorant with evaporating water during drying, causing unlevelness.
- **Curing:** heat treatment crosslinking the binder film.
- **Antimigrant:** viscosity-raising auxiliary that suppresses pigment migration.

7. Key points

- Depth = concentration × pick-up; permanence = binder cure.
- Drying, not padding, is where pigment dyeing succeeds or fails (migration).
- Judge pigment dyeing on rub fastness and handle, not on wash fastness alone.

8. Summary

Pigment dyeing exchanges chemical affinity for mechanical filming: a padder meters colour, controlled drying keeps it where it was put, and curing locks it in a polymer skin. Its blend-uniformity and economy are unmatched; its surface character is its signature limitation — the same trade-off that printing exploits in Topic 10.

9. Practice

1. A liquor of 20 g/L pigment is padded at 60% pick-up. Calculate the pigment add-on in % owf. The buyer wants 1.8% owf at the same pick-up — what liquor concentration is needed?
2. A pigment-dyed fabric is deeper on the face than the back. Which step is responsible, and what two corrections will you apply?
3. Why does doubling binder improve rub fastness but worsen handle?

Answer key

1. Add-on = 20 × 0.60/10 = 1.2% owf (20 g/L × 0.60 L/kg = 12 g/kg). For 1.8% owf: 18 g/kg ÷ 0.60 L/kg = 30 g/L.
2. Drying migration — liquor moved to the face evaporation front. Apply antimigrant in the liquor and IR pre-drying (remove ~30% moisture contactlessly) before hot-flue drying; also balance air flow

both faces.

3. Rub fastness fails when pigment particles at the film surface are abraded away; more binder buries particles deeper in a continuous tougher film. The same extra film is a thicker polymer coating, which the hand feels as stiffness.
